GG-M-112a
15 MAY 1951
SUPERSEDING
Fed. Spec. GG-M-112
12 MAY 1947

FEDERAL SPECIFICATION

MANOMETER OUTFITS, SPINAL-FLUID (WATER-TYPE)

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types and classes. — Spinal-fluid (water) manometers in accordance with this specification shall be of one type and the following classes, as specified in the invitation for bids:

Class A.—Without case.
Class B.—With wood case.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

GG-I-526—Instruments, Dental and Surgical; General Specifications.
GG-N-196—Needles, Hypodermic; for Luer Syringes.
GG-S-921—Syringes; All-Glass, Luer.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

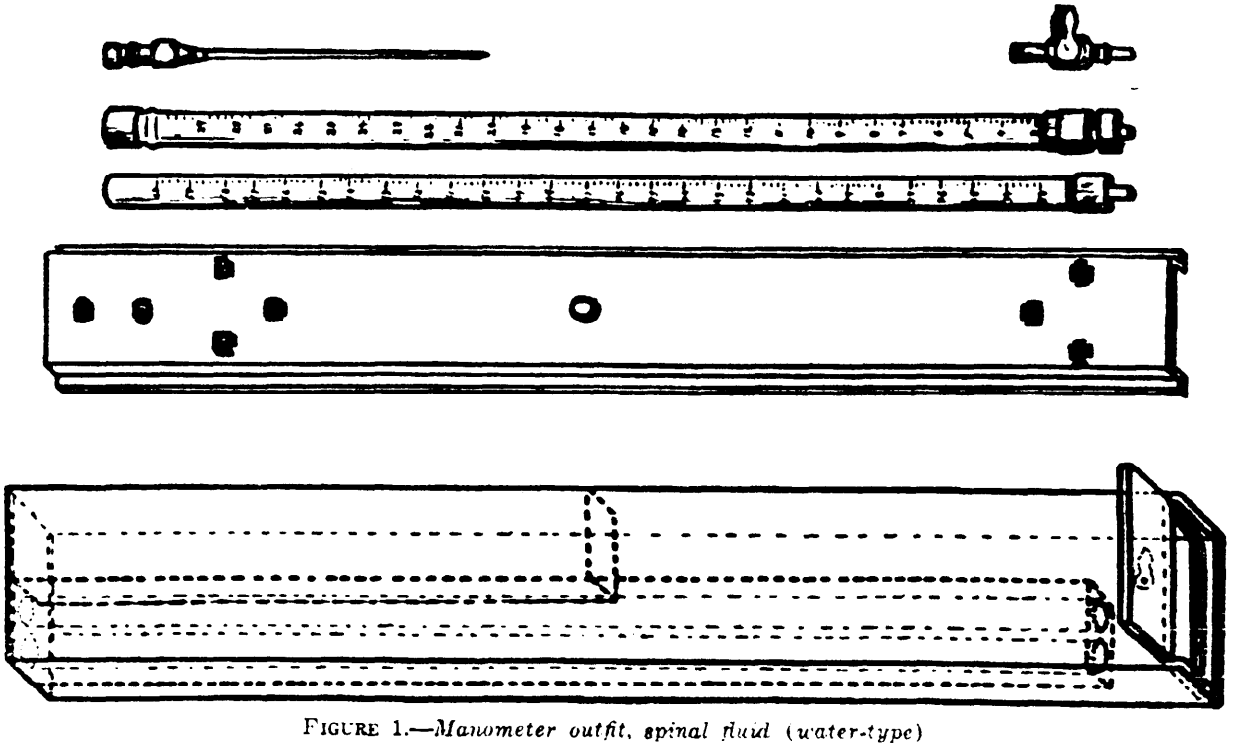


FIGURE 1.—*Manometer outfit, spinal fluid (water-type)*

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2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—The glass for capillary tubes shall be as specified hereinafter.

3.1.2 *Brass.*—The brass shall conform to the requirements of Federal Specification GG–I–526, class 4, composition N.

3.1.3 *Steel.*—The steel shall conform to the applicable requirements of Federal Specification GG–N–196 for type I needles.

3.1.4 *Wood.*—The wood for the case shall be hardwood.

3.2 Design.—The spinal-fluid manometer outfits shall be of the general design illustrated in figure 1. The outfit proper shall consist essentially of two capillary tubes, a spinal needle, a three-way stopcock, and a metal sterilizing tray. The general design of the case used for class B outfits is illustrated also in figure 1. Standard Luer tips shall be in accordance with Federal Specification GG–S–921; standard Luer cavities shall be in accordance with Federal Specification GG–N–196.

3.3 Capillary tubes.—The tubes, one lower and one upper, shall be of glass. The inside diameter shall be 1.5 ± 0.25 millimeters and the outside diameter 7.0 ± 1.0 millimeters. The lower and upper tubes shall be interchangeable.

3.3.1 *Lower tube.*—The lower tube shall be fitted at the lower end with an adapter with standard Luer tip that will make a nonleakable connection with the Luer cavities of the stopcock. The upper end shall be fitted with an adapter for connection with the upper tube; the free end of the adapter shall have a standard Luer cavity.

3.3.2 *Upper tube.*—The lower end of the upper tube shall be provided with an adapter for connection with the lower tube; the free end of the adapter shall have a standard Luer tip which shall make a nonleakable connection with the lower tube.

3.3.3 *Graduation scale.*—The tubes shall be graduated from 4 centimeters (40 mm.) or lower to not less than 60 centimeters (600 mm). The scale on the glass tubes shall be marked in 2 millimeter graduations with main graduations every 1 centimeter (10 mm). The main graduations shall be not less than 3 millimeters in length. Figures indicating the number of centimeters shall appear either at each main graduation or at alternate main graduations. The subgraduations shall be plainly distinguished from the main graduations and shall be not less than 2 millimeters in length. The numerals and graduation markings shall be clearly defined, legible, and shall be of such material and application that they will withstand the test for permanency of marking described in 4.3.4 The marking on tested samples shall show no significant change from those samples not subjected to the test. The graduation markings shall be consecutive when the upper and lower tubes are joined. Connection of the two tubes may obliterate the glass tube graduations for not more than 5 centimeters.

3.3.4 *Solubility.* — No pink color shall develop in the testing solution when the tubes are tested as described in paragraph 4.3.1.1.

3.3.5 *Thermal shock.* — The tubes shall withstand, without breakage, the test described in paragraph 4.3.1.2.

3.4 **Spinal puncture needle.**—The needle shall be 20-gage, 3.50 inches long, and shall conform to requirements of Federal Specification GG–N–196 for type I needles, style b short bevel point, style 2 hub. Stylet shall be of the conventional type for spinal puncture needles.

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3.5 **Three-way stopcock.** — The stopcock shall be of brass. It shall be capable of directing the flow of the spinal fluid from the spinal puncture needle either upward at a right angle into the manometer tube or horizontally into a syringe, test tube, etc., or of shutting off the flow. A tip with a standard Luer taper conforming to the applicable requirements of Federal Specification GG-S-921 shall make a nonleakable union with the spinal puncture needle; the other two openings shall be standard Luer cavities conforming to the applicable requirements of Federal Specification GG-N-196. The stopcock and first adapter may take the column fluid to the graduation at 3 centimeters. The valve directing the flow shall be readily and easily controlled in such a manner that the three operations of the stopcock are accomplished within one half-turn of the controlling handle.

3.6 **Sterilizing tray.**—The sterilizing tray shall be made of hard sheet brass, chromium plated or nickel plated. The brass shall be of sufficient thickness and so constructed that it will not bend under ordinary use. The tray shall be fitted with permanently attached metal clasps, spring clamps, or other similar means to hold securely all parts of the outfit. A metal handle, 1 inch high, shall be attached at the center of the tray.

3.6.1 *Dimensions.* — The sterilizing tray shall be approximately 13.5 inches long by 1.0 inch wide, to fit properly within a case approximately 14.0 inches long and 2.5 inches wide.

3.7 **Plating.**—The plated metal parts shall conform to the applicable requirements of Federal Specification GG-I-526, except that a chromium coating is not required.

3.8 **Marking.**—Unless otherwise specified, glass parts and stopcock shall be permanently marked with the name or registered trademark of the manufacturer.

3.9 **Instruction sheet.**—A complete set of instructions for the use and care of the manometer outfit shall be supplied with each apparatus.

3.10 **Class A.**—Class A manometer outfits shall be furnished without a case.

3.11 **Class B.**—Class B manometer outfits shall be provided with a hardwood case (fig. 1), approximately 14.0 inches long, 2.5 inches wide, and 1.75 inches deep. Separate spaces shall be provided within the case for each manometer tube, the tray, and the other parts.

3.11.1 *Finish.*—The case shall have a finish of not less than one coat of stain and one coat of lacquer.

3.11.2 *Case marking.*—The outside of case shall be legibly marked with the name or registered trade-mark of the manufacturer.

3.12 **Workmanship.** — The workmanship shall be first class throughout. The manometer outfits shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—At least two manometer outfits of each class shall be taken at random from each delivery, for inspection and tests.

4.2 **Inspection.**—Samples shall be inspected for material, workmanship, design, fit of parts, dimensions, construction, finish, packaging, packing, and marking.

4.3 **Tests.**

4.3.1 *Glass tubes.*

4.3.1.1 *Solubility.* — The glass capillary tubes shall be washed and rinsed in distilled water at room temperature for 10 minutes. They shall then be partially filled to one-fourth of their capacity with CO_2-free dis-

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shall be examined for evidence of leakage.

tilled water containing 0.2 percent U. S. P. standard phenolphthalein test solution and autoclaved at 15 pounds pressure for 30 minutes at not less than 248° F.

4.3.1.2 *Thermal shock test.* — The glass capillary tubes shall be placed in an oven which has been previously heated to 100° ± 1° C., and allowed to remain therein at this temperature for at least 30 minutes. They shall then be removed from the oven and immediately plunged into water at a temperature of 5° C.

4.3.2 *Needles.*—Needles shall be tested to determine conformance to the requirements of Federal Specification GG–N–196.

4.3.3 *Plating.* — The plated metal parts shall be tested to determine conformance to the requirements of Federal Specification GG–I–526.

4.3.4 *Permanency of marking.*—Samples shall be tested for permanency of marking by the following method:

4.3.4.1 *Alkali test.* — Immerse specimens completely in sodium phosphate (dibasic) 0.5 normal solution or sodium phosphate (tribasic) 0.3 normal solution or sodium carbonate 0.5 normal solution, at room temperature for 18 hours.

4.3.4.2 *Acid test (sulphuric).* — Immerse specimens completely in concentrated sulphuric acid at room temperature. Let stand 18 hours.

4.3.4.3 At the end of each test period remove specimens and rinse thoroughly in distilled water. There should be no evidence of color in the test solution and no evidence of removal of the markings from test specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Class A.*—Unless otherwise specified, each spinal-fluid manometer outfit, properly mounted on the tray, shall be packaged in a sturdy cardboard box of appropriate size, reinforced with backing if necessary, and suitable for use as a permanent case.

5.1.2 *Class B.* — All components of the spinal-fluid manometer outfit shall be contained in its wooden case. The case shall be packaged in a cardboard container.

5.2 Packing.—The manometer outfits shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Each cardboard box or carton shall be plainly labeled with the name of the item, class, quantity contained, and the name of the manufacturer or contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and class of item as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Purchasers should specify class required and should exercise any desired options offered herein. (See pars. 1.1, 3.8, 5.1.1, 5.3.2, 6.2, and 7.2.6.).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with the requirements of this specification. If, for any particular purpose, samples with

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bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type and classes which are normally procured for Federal Government requirements.

6.4 This specification covers the following item listed in the Armed Services Catalog of Medical Matériel:

3–469–200 Manometer, Spinal, Water Type.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications.*

Federal Specifications:

NN–B–621 —Boxes, Wood, Nailed and Lock-Corner.

LLL–B–631 —Boxes, Fiber, Corrugated (for Domestic Shipment).

LLL–B–636 —Boxes, Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN–P–106 —Packaging and Packing for Overseas Shipment — Boxes, Wood, Nailed.

JAN–P–125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

Jan–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case Liner.

MIL–M–3157—Marking of Domestic Shipments by Contractors for Supplies and Equipment Consigned to the Medical Departments of the Armed Services.

Navy Department Specifications:

General Specifications for Inspection of Material.[1]

(Copies of specifications required for Army and Air Force purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and Military specifications required for Navy purchases and Navy Department specifications may be obtained without cost upon application to Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, New York. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.2.2 *Inspection procedures.* — For Naval purchases, the general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.3 *Packaging.*—One spinal manometer outfit constitutes one unit. Each unit shall be packaged as specified in 5.1.1 or 5.1.2. Fragile items shall be properly protected by sufficient cushioning material (kimpack,

[1] Applicable to Navy purchases only.

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cellulose, die-cut pads, cells, etc.) to insure against shifting, breakage, or damage. Each unit shall be further packaged in a carton of appropriate size conforming to the requirements of Federal Specification LLL-B-631 or LLL-B-636. Each carton shall be adequately secured to prevent accidental opening.

7.2.4 *Packing.*

7.2.4.1 *Domestic.*—Unless otherwise specified, 24 packages (24 units) shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-621.

7.2.4.2 *Overseas.*—Unless otherwise specified, 24 packages (24 units) shall be packed in a shipping container designed for type 3 load and conforming to the requirements of Military Specification JAN-P-106. Closure and strapping shall conform to the appendix thereto. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made of material conforming to the requirements of Military Specification JAN-P-125, for case liners; seams and closures shall be sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification JAN-P-140.

7.2.4.3 *Packing variation permitted.* — If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design, so constructed as to insure safe delivery to specified destination.

7.2.5 *Marking.*

7.2.5.1 *Packages.*—Each package shall be marked with the stock number, nomenclature, unit, quantity, and the name of the contractor. When the contractor is not the manufacturer, the name of the manufacturer shall also appear as the manufacturer.

7.2.5.2 *Shipping containers.*

7.2.5.2.1 *Domestic.* — Each shipping container shall be marked in accordance with Military Specification MIL-M-3157.

7.2.5.2.2 *Overseas.* — Each shipping container shall be marked as specified by the procuring agency.

7.2.6 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify domestic or overseas packing and marking of shipping containers. (See 7.2.4 and 7.2.5.2).

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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